

Instruction for cell voltage offsets in EEPROM

This is mostly relevant for the cell imbalance indicator. From my experience the ISL has a systematic error of ± 30 mV across the cells. When you balance your cells externally and your multimeter shows all cells to have exactly the same voltage you will likely find your battery pack to report an imbalance between 0 mV and 60mV although it is actually 0 mV. This is just part variance. The imbalance indicator will only be reliable for imbalance greater e.g. 100 mV. The step of determining an setting the voltage offsets is completely optional though.

Procedure to correctly set the offset voltage offsets:

- 1) Programm the PIC with the firmware
- 2) Disconnect the battery pack from Pickit and let it idle until it goes to sleep. (This will write the currently measured cell voltages to EEPROM)
- 3) Reconnect the battery pack to Pickit and use the Pickit programming tool to read the EEPROM.
- 4) Compare the cell voltages saved to EEPROM to the voltages you measure with your multimeter
- 5) Calculate the necessary voltage offsets and save them to the appropriate locations in the EEPROM. Keep in mind that the currently set voltage offsets are already applied in the reported cell voltages.
- 6) Repeat the steps 2-5 until the reported voltages match the measured values ± 1 mV.

The default values for the voltage offsets are +1 mV.

These values are written to EEPROM during programming (or manually) to the following locations:

0x48: offset voltage cell 1 / mV

0x49: offset voltage cell 2 / mV

0x4A: offset voltage cell 3 / mV

0x4B: offset voltage cell 4 / mV

0x4C: offset voltage cell 5 / mV

0x4D: offset voltage cell 6 / mV

Note that the stored values are interpreted as signed 8 bit integers. Use a hex-decimal converter.

E.g. a value of 1A stored in the location 0x48 will result in an offset voltage of +26 mV for cell 1. A value of E6 will give an offset of -26 mV.

The reported voltages by the battery pack are stored in the locations:

0x38-0x39: voltage cell 1 / mV

0x3A-0x3B: voltage cell 2 / mV

0x3C-0x3D: voltage cell 3 / mV

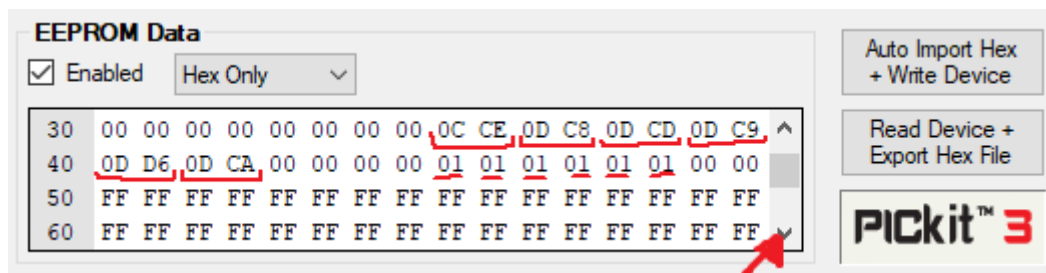
0x3E-0x3F: voltage cell 4 / mV

0x40-0x41: voltage cell 5 / mV

0x42-0x43: voltage cell 6 / mV

These values are unsigned 16 bit values and are represented via two subsequent 8 bit values. Use a hex-decimal converter.

E.g. a value of 0CCE stored in the location 0x38 and 0x39 will result in a reported cell voltage 1 of 3278 mV.



In the Pickit programming tool scroll down three lines to match this screenshot.